

Polarisation and scattering — every combination, and how to choose

SAR reference sheet · Earth Observation · V. Emberumal 2248220 | Grey rows with a thick rule = what Sentinel-1 gives you for free

1 THE FOUR LINEAR CHANNELS — FIRST LETTER TRANSMITS, SECOND RECEIVES

CHANNEL	NAME	PHYSICALLY SENSITIVE TO	TYPICAL BACKSCATTER, FOREST	WHEN YOU WANT IT
VV	co-polarised	Bragg surface scattering. Vertical structures: stems, stalks, ripples on water.	−7 to −9 dB	Soil moisture, open water, oil slicks, crops with vertical stems.
HH	co-polarised	Double-bounce from dihedrals: trunk–ground, wall–street, ice ridges.	−6 to −8 dB	Flooded forest, urban, sea ice, ship detection in co-pol.
VH	cross-polarised	Depolarisation. Requires multiple scattering — a randomly oriented volume.	−14 to −17 dB	Biomass, canopy volume, crop growth stage, land–water separation in wind.
HV	cross-polarised	Identical physics. For a monostatic radar, reciprocity gives HV = VH.	−14 to −17 dB	Same as VH. Which one you get depends on the mission's transmit sequence.

The rule to remember: a single reflection cannot rotate the polarisation plane. Any cross-polarised return is proof that the wave bounced more than once — which is why VH is the volume channel.

2 THE FOUR CANONICAL SCATTERING MECHANISMS AND THEIR POLARIMETRIC SIGNATURE

MECHANISM	GEOMETRY	DOMINANT CHANNEL	HH/VV RATIO	∠(HH–VV) PHASE	ALPHA ANGLE	REAL TARGETS
Surface (single bounce, Bragg)	one reflection off a rough plane	VV > HH	< 1	≈ 0°	≈ 0°	Bare soil, calm water, roads, sand, burnt bare ground
Double bounce (dihedral)	two reflections at 90°	HH > VV	> 1	≈ 180°	≈ 90°	Buildings, flooded forest trunks, ship hulls, tree–ground
Volume (random dipoles)	many reflections inside a canopy	HV = VH high	≈ 1	random	≈ 45°	Forest canopy, dense crops, dry snow, vegetated dunes
Helix / oriented	non-symmetric, man-made	circular components	—	—	—	Complex urban, rotated dihedrals, some ships

For an ideal random volume of thin dipoles the theoretical ratio is $\sigma_{HV}/\sigma_{HH} = 1/3$ (≈ −4.8 dB). A measured cross-pol ratio far above that means oriented structure, not a random volume.

3 WAVELENGTH — WHAT THE WAVE CAN ACTUALLY REACH

BAND	λ	PENETRATES TO	DOMINANT INTERACTION IN FOREST	BIOMASS SATURATION	MISSIONS
X	≈ 3 cm	leaf surface, top few cm of canopy	leaves, twigs — a 'skin' image	≈ 10–30 t/ha	TerraSAR-X, COSMO-SkyMed
C	≈ 5.6 cm	upper canopy, small branches	leaves and small branches; saturates early	≈ 20–50 t/ha	Sentinel-1, RADARSAT-2, RCM
L	≈ 24 cm	through canopy to branches, trunks, some soil	branch and trunk scattering, trunk–ground double bounce	≈ 100–150 t/ha	ALOS-2, SAOCOM, NISAR, ROSE-L
P	≈ 70 cm	trunks and soil; deepest of all	trunk–ground, direct trunk	≈ 200–300 t/ha	BIOMASS (ESA)

4 PRACTICAL SELECTION — THE APPLICATION DECIDES THE POLARISATION

APPLICATION	BEST CHOICE	WHY, IN ONE LINE	TRAP TO AVOID
Open-water flood mapping	VH, with VV as check	Smooth water is a specular reflector: almost no return, so it reads black against land.	Wind roughens the surface and brightens VV; VH is far less affected.
Flood under forest	HH, L-band	Trunk–water dihedral makes the flooded stand brighter, not darker.	C-band cannot get through the canopy. This is an L-band question.
Above-ground biomass	HV, L or P-band	Cross-pol scales with canopy volume until it saturates.	C-band VH saturates by ≈ 50 t/ha — useless for mature forest.
Crop type & phenology	VV + VH time series	The VH/VV ratio tracks the shift from soil to canopy dominance.	Single-date classification. The information is in the temporal shape.
Soil moisture	VV, low incidence, L-band	Bragg backscatter rises with the dielectric constant of wet soil.	Vegetation and roughness both mimic a moisture signal.
Urban / built-up	HH	Wall–ground dihedrals give a strong, stable double bounce.	Streets rotated away from the look direction lose their double bounce entirely.
Ship detection	HV / VH	Ocean cross-pol clutter is very low, so metal targets stand out.	Low absolute signal — thermal noise removal is mandatory.
Oil spill	VV	Oil damps the capillary waves that create the Bragg return → a dark slick.	Low wind, biogenic slicks and rain cells are look-alikes.
Sea ice	HH + HV (EW mode)	HH is less incidence-angle dependent over ice; cross-pol separates ice types.	Wet ice surfaces collapse the contrast.
Burn severity / structure	VH change + InSAR coherence	Losing the canopy removes volume scattering, so VH should fall.	This is what failed in my project. Amplitude is confounded by slope and moisture. Coherence is the robust observable.
Deformation / subsidence	any co-pol, but same track	Phase, not amplitude. Polarisation barely matters; geometry is everything.	Mixing ascending and descending passes. Never do it.

5 POLARIMETRIC DECOMPOSITIONS — AND WHAT DATA EACH ONE DEMANDS

DECOMPOSITION	TYPE	WHAT IT RETURNS	DATA NEEDED	AVAILABLE FROM SENTINEL-1?
Pauli	coherent	RGB: R = [HH–VV] double bounce, G = [HV] volume, B = [HH+VV] surface	full quad-pol, complex	No — needs all four channels
Freeman–Durden	incoherent, model-based	Power split into surface + double-bounce + volume	quad-pol covariance	No
Yamaguchi	incoherent, 4-component	Adds a helix term for complex urban scenes	quad-pol	No
Cloude–Pottier	incoherent, eigenvector	Entropy H, Anisotropy A, mean alpha α — the H/α plane	quad-pol (full form)	No, only a dual-pol H/α
Dual-pol H/α	incoherent, 2×2	Reduced entropy and alpha from the VV+VH covariance	dual-pol complex (SLC)	Yes — from SLC, not GRD
Ratio & RVI	simple indices	VH/VV; RVI = 4σ _{VH} /(σ _{VV} +σ _{VH})	dual-pol intensity only	Yes — works on GRD

THE SENTINEL-1 LIMIT, STATED PRECISELY

Sentinel-1 is **dual-polarised, never quad-polarised**. Over land in IW mode it transmits V and receives both V and H, giving **VV + VH**. In the polar regions and in EW mode it switches to **HH + HV**.

Consequences you must be able to state: the scattering matrix is incomplete, so the **co-pol phase difference ∠(HH–VV) is not measurable**, and with it go Pauli, Freeman–Durden and the full H/α plane. What remains is the dual-pol covariance, the **VH/VV** ratio, RVI, and — the strongest of them — **interferometric coherence**.

INCIDENCE ANGLE — THE VARIABLE EVERYONE FORGETS

Backscatter is a function of **local** incidence angle, not the nominal one. Shallow angles favour surface scattering and exaggerate topography; steep angles let volume scattering dominate. Cross-pol is the least angle-dependent of the four channels, which is one more reason it travels well between tracks.

On slopes, σ° must be replaced by **Υ°_T** through radiometric terrain flattening before any two dates or tracks are compared. Skipping that step is what produced the physically impossible **+1.97 dB after the fire** in my own Sentinel-1 check; separating ascending from descending removed 77 % of it.

6 ACQUISITION MODES YOU WILL BE ASKED TO NAME

MODE	SWATH × RESOLUTION	DEFAULT POLARISATION	USED FOR
IW — Interferometric Wide (TOPS)	250 km × 5 × 20 m	VV+VH over land	The land default. Everything in this project.
EW — Extra Wide	400 km × 20 × 40 m	HH+HV	Sea ice, polar oceans, maritime surveillance.
SM — Stripmap	80 km × 5 × 5 m	selectable dual	Emergency tasking, small islands.
WV — Wave	20 × 20 km vignettes	VV	Ocean wave spectra over open sea.

One-line answer if asked “why didn’t you just use polarimetry?” — Because a full polarimetric decomposition needs the complete scattering matrix, and no free, systematically acquired, global mission provides quad-pol. Sentinel-1 gives two channels; the physics I could extract from them is the cross-pol ratio and the interferometric coherence, and of those, coherence is the one that answers a structural question without depending on radiometric calibration at all.